N-T-101e
June 20, 1967
SUPERSEDING
Int. Fed. Spec. N-T-00101d (Army-GL)
November 3, 1966 and
Fed. Spec. N-T-101c
October 22, 1963

FEDERAL SPECIFICATION

TAPIOCA

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for two types of tapioca for use by all Federal agencies.

1.2 Classification.

1.2.1 Types and sizes. Tapioca covered by this specification shall be of the following types and sizes, as specified (see 6.1).

Type I - Pearl.

Size 1 - Medium.
Size 2 - Small.

Type II - Granulated.

Size 1 - Coarse.
Size 2 - Medium.
Size 3 - Small.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

RR-S-366 - Sieve, Test.
PPP-B-636 - Box, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Specification:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

Governmental:

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Copies may be obtained upon application, acceompanied by money order, coupon, or cash, to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402. Prices may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Material. Tapioca shall be prepared from the starch obtained from the tuberous roots of the cassava plant also known as manioc. The starch obtained from the root of the cassava or manioc shall be white, clean, and free from evidence of insect infestation, other foreign matter, off odors, and off flavors.

3.2 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.3 Finished product. The finished product shall be bright, white, clean, free flowing, and free from dirt, grit, other foreign matter, or any off flavors and odors.

3.3.1 Granulation.

3.3.1.1 Type I, size 1, pearl, medium. Type I, size 1, shall be pearls, all of which shall pass through a U. S. Standard 1/4-inch sieve and not less than 90 percent by weight shall be retained on a U. S. Standard No. 4 sieve. Type I, size 1, shall contain not more than 1 percent broken pearls by weight.

3.3.1.2 Type I, size 2, pearl, small. Type I, size 2, shall be pearls (of practically uniform size) all of which shall pass through a U.S. Standard No. 6 sieve. Type I, size 2, shall contain not more than 1 percent broken pearls by weight.

3.3.1.3 Type II, size 1, granulated, coarse. Type II, size 1, shall be granules, all of which shall pass through a U. S. Standard No. 10 sieve and not less than 95 percent by weight shall be retained on a U. S. Standard No. 20 sieve.

3.3.1.4 Type II, size 2, granulated, medium. Type II, size 2, shall be granules, all of which shall pass through a U. S. Standard No. 10 sieve and not less than 60 percent by weight shall be retained on a U. S. Standard No. 20 sieve.

3.3.1.5 Type II, size 3, granulated, small. Type II, size 3, shall be granules, all of which shall pass through a U. S. Standard No. 20 sieve.

3.4 Workmanship.

3.4.1 For civil agencies. The processing plant, grounds, equipment, packaging materials, personnel practices, and sanitary practices used in the production of this item, shall be such as to minimize the possibility of contamination of the product through microbial growth, dust, pests, rodents, condensate, or other unsanitary sources. The product shall be prepared and processed with minimum delay between the various stages of production.

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3.4.2 For military agencies. The product shall be processed in establishments meeting the requirements of MIL-STD-668. It shall be prepared and processed with minimum delay between the various stages in production.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection.

4.1.1 General. Inspection shall be made by the procuring agency or a duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing and packaging, at any suitable point in transit, and after delivery to the point of destination.

4.1.2 Factory inspection. The passing as satisfactory of any detail of processing or materials shall not relieve the supplier of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.1.3 Final inspection. Unless otherwise specified, final inspection shall be made after delivery to the point of destination.

4.2 Responsibility for inspection. The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to perform the examinations and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

4.2.1 Inspection records of the examinations and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2.2 Sampling. Samples of finished products covered by this specification and samples of any materials, components, or constituents entering products covered by this specification may be taken at any time by the Government inspector to determine compliance with the requirements. Such samples as are required shall be at the expense of the supplier.

4.2.3 Preacceptance inspection (plant sanitation inspection).

4.2.4 For civil agencies. Inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions on which have received prior sanitary approval recommended by the inspection agency.

4.2.5 For military agencies. Award of contracts shall be limited to establishments which have received prior sanitary approval by the Military Veterinary Services in accordance with MIL-STD-668.

4.2.6 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated herein.

4.2.7 Component and material inspection. In accordance with 4.2.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.8 Examination of starch. Conformance of starch to identity requirements of 3.1 shall be ascertained by examination of pertinent labels, invoices, or other valid documents. In addition, the starch shall be examined organoleptically, as necessary, to determine conformance to condition requirements of 3.1. The sample unit for examination shall be 1 pound taken from each of 5 primary containers in the lot. If the lot consists of less than 5 containers, a sample unit shall be extracted from each container in the lot. Nonconformance of the starch to one or more requirements shall be indicative of its unacceptability and its use shall be cause for rejection of the involved quantity of the end item.

4.2.9 Examination of end item. Classification of defects found during examination shall be in accordance with tables II through IV and 4.2.10, 4.2.11, and 4.2.12. The sampling plans and acceptance criteria shall be as shown in table I. The AQLs for tables II and IV shall be expressed in terms of defects per hundred units and in terms of percent defective, for table III.

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TABLE I. End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLs			
				Major A	Major B	Minor	Total
II	I	Primary container	Primary containers	0.25	1.5	-------	4.0
III	S-4	Primary container	Primary containers	-------	-------	2.5	------
IV	S-2	Contents of 1 primary container	Primary containers	-------	2.5	-------	------

TABLE II. External examination of primary container

Category			Defect
Major A	Major B	Minor	
101			Not closed.
	151		Sifter, tear, or hole.
	152		Foreign or off odor.
	153		Nomenclature missing, incorrect, or illegible.
		201	Labeling information other than nomenclautre missing, incorrect, or illegible.
		202	Not clean.

TABLE III. Examination for net weight 1/ 2/

Category	Defect
Minor	
201	Net weight more than 5 percent under specified weight (container required to have a net weight of 1 pound or less).

1/ Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight.

2/ Weights shall be recorded to nearest 1/4 ounce.

TABLE IV. Examination of product 1/ 2/

Category	Defect
Major B	
	Pearl, medium, type I, size 1
151	Less than 100 per cent passes through a U. S. Standard 1/4 inch sieve.
152	Less than 90 percent by weight retained on a U. S. Standard No. 4 sieve.
153	More than 1 percent by weight broken pearls.

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TABLE IV. Examination of product (cont'd.)

Category	Defect
Major B (cont'd.)	
	Pearl, small, type I, size 2
154	Not practically uniform in size.
155	Less than 100 percent passes through a U. S. Standard No. 6 sieve.
156	More than 1 percent by weight broken pearls.
	Granulated, coarse, type II, size 1
157	Less than 100 percent passes through a U. S. Standard No. 10 sieve.
158	Less than 95 percent by weight retained on a U. S. Standard No. 20 sieve.
	Granulated, medium, type II, size 2
159	Less than 100 percent passes through a U. S. Standard No. 10 sieve.
160	Less than 60 percent by weight retained on a U. S. Standard No. 20 sieve.
	Granulated, small, type II, size 3
161	Less than 100 percent passes through a U. S. Standard No. 20 sieve.

1/ Presence of foreign material (e.g., rust, dirt, grease, particles of wood, paint, glass, or metal, hair, insects, insect fragments, larvae, etc.) shall be cause for rejection of the entire lot.

2/ Sieve test data shall be recorded to the nearest 1 percent.

4.2.10 Examination of shipping container (for civil agencies). The shipping container shall be examined to determine compliance to marking requirements of 5.3.1.2.

4.2.11 Examination of shipping container. Examination of the filled and closed shipping container for levels A and B packing shall be in accordance with the examination criteria of PPP-B-636. In addition, the following defects shall be included in the table of examination: Major: markings missing, incorrect, or illegible; Minor: arrangement not as specified. For level C packing, examination shall be performed in accordance with the aforementioned specification except that only the defects for marking and arrangement shall apply.

4.2.12 Examination of waterproof case liner. When waterproof case liners are required to be in accordance with MIL-L-10547, examination of filled and sealed waterproof case liners shall be in accordance with that specification.

4.2.13 Standby test samples. The Government reserves the right to withdraw and hold standby test samples of components or finished product or both (quantity of which shall be not more than twice that of the original Government sample) for inspection purposes. These samples, if not used, will be returned to the vendor.

4.3 Tests. Test procedures and controls which differ from those specified herein may be used by the supplier, if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

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4.3.1 Sieving test. U. S. Standard woven-wire-cloth sieves which are 8 inches in diameter, which have full-height frames, and which comply with RR-S-366, shall be used in making the sieving test. The sieves shall be fitted together in the following order:

Type I, size 1 test -	Cover
	U. S. 1/4 inch
	U. S. No. 4
	Pan
Type I, size 2 test -	Cover
	U. S. No. 6
	Pan
Type II tests -	Cover
	U. S. No. 10
	U. S. No. 20
	Pan

4.3.1.1 Procedure. Place a 100-gram sample on the top sieve of an assembled set of specified sieves, including cover and catch pan. Place the sieve assembly on a Tyler Rotap, or similar type machine, and sift for five minutes. Report percent passing through or retained, as applicable. If the procedure is carried out by hand, with the left hand, hold the sieve assembly at an agle of about 30° from the horizontal, so that the upper edge of the tilted plane is toward the right hand. Shake by striking the right side of the sieve assembly with a short, sharp stroke with the right hand moving in a horizontal plane. Strike at the rate of about 150 times per minute. After each 25 strokes, turn the sieve assembly 1/16 of a revolution in the same direction. Continue shaking for 2 minutes. Remove, weigh, and calculate the percentage of the sample, segregated by the sieving action.

5. PREPARATION FOR DELIVERY

5.1 Packaging, all levels. The product shall be packed in accordance with 5.1.1 or 5.1.2, as applicable.

5.1.1 Civil agencies. The product shall be packaged in the unit quantity specified in the contract or order in accordance with good commercial practice for the commodity.

5.1.2 Military agencies. Eight ounces of the type II product shall be packaged in a commercial carton of the type normally used for the product.

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Twenty-four pounds of the product shall be packed in a snug-fitting fiberboard box, constructed, closed, and reinforced in accordance with style RSC, V2s of PPP-B-636. The shipping containers shall be provided with a waterproof case liner fabricated and sealed in accordance with type I, II, or III, grade C, for subsistence items, of MIL-L-10547.

5.2.2 Level B. Twenty-four pounds of the product shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with type CF or SF, class domestic, method II closure of PPP-B-636.

5.2.2.1 When specified (see 6.1 and 6.2), the product shall be packed in accordance with 5.2.1, except that the shipping container shall be constructed in accordance with style RSC, V3s or V3c of PPP-B-636.

5.2.3 Level C. The product shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the rules or regulations of the carrier applicable to the mode of transportation.

5.3 Labeling and marking.

5.3.1 Civil agencies.

5.3.1.1 Unit containers. Any commercial labeling and additional labeling as specified, that complies with the Federal Food, Drug, and cosmetic Act, and Regulations Promulgated Thereunder, is acceptable.

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5.3.1.2 Shipping containers. Shipping containers shall be marked in accordance with the requirements of Fed. Std. No. 123, and such other information as specified by individual agencies.

5.3.2 Military agencies.

5.3.2.1 Unit containers. The unit container shall be printed or labeled clearly and legibly and shall include the name of the product, the name of the manufacturer, the net weight of the product, and any other information required by the Federal Food, Drug, and Cosmetic Act, and Regulations Promulgated Thereunder.

5.3.2.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size of product required (see 1.2).
(c) Packaging and packing levels (see 5.1, 5.2, and 5.2.2.1).

6.2 Packaging specified in 5.2.2.1 is intended to provide the limited military protection required for: shipment of supplies to certain specified overseas destinations, transfer at sea, and shipment and storage of Special Program Reserves.

6.3 If samples with bids are necessary, they should be specifically asked for in the invitation for bids, or request for proposal, and the particular purpose to be served by the bid samples should be definitely stated.

6.4 Supplier's inspection. The supplier should be required to assure that the product conforms to the specification prior to submission to the Government for final inspection.

6.5 Destination inspection. When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

MILITARY CUSTODIAN:

Army - GL
Navy - SA
Air Force - 45

Preparing activity:

Army - GL

Review activities:

Army - MD, GL
Navy - MC, SA
Air Force - 45

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

U. S. Government Printing Office: 1967

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Tapioca - N-T-101e

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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